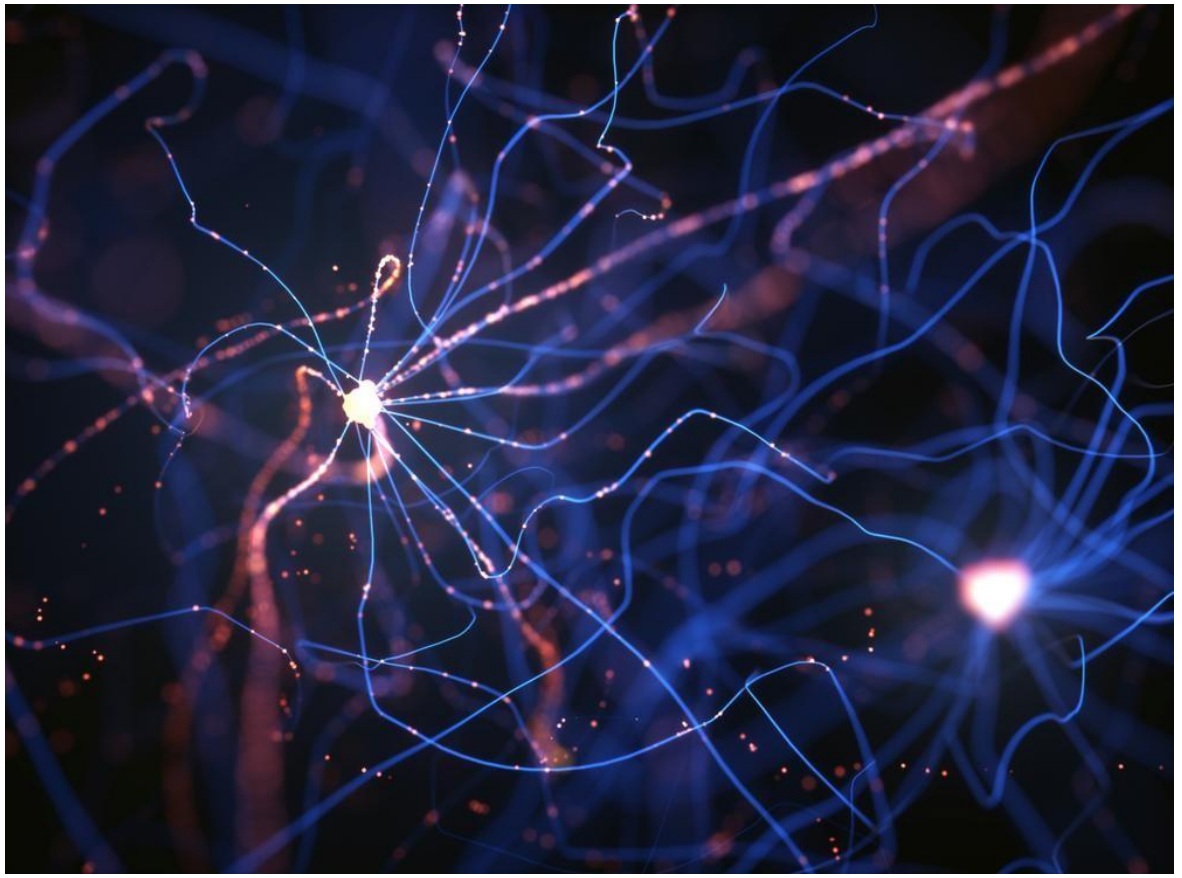
Deep Learning

NIELIT ROPAR

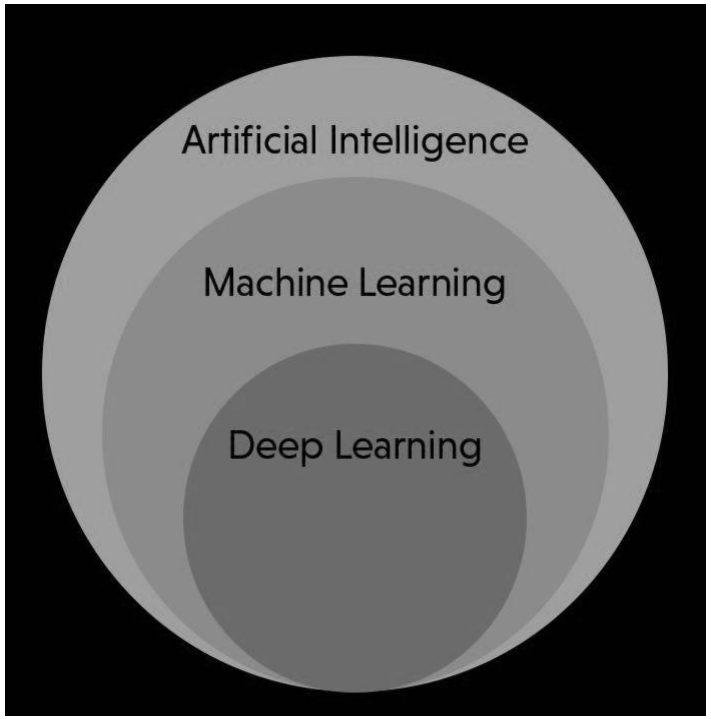
Fundamentals of Deep Learning



What is Deep Learning?

"Deep Learning doesn't do different things,
it does things differently"

Deep Learning is a type of Machine Learning



Machine Learning vs. Deep Learning

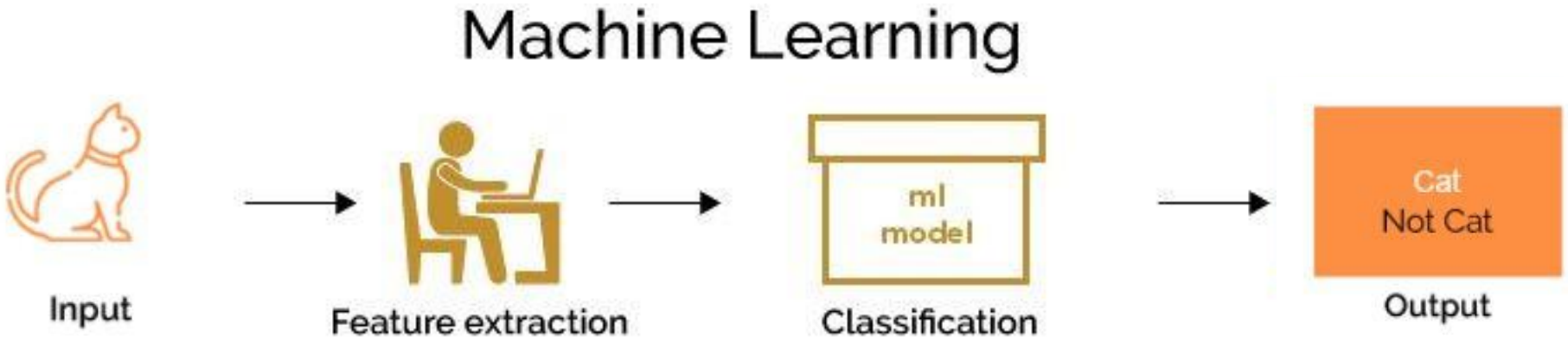


Input



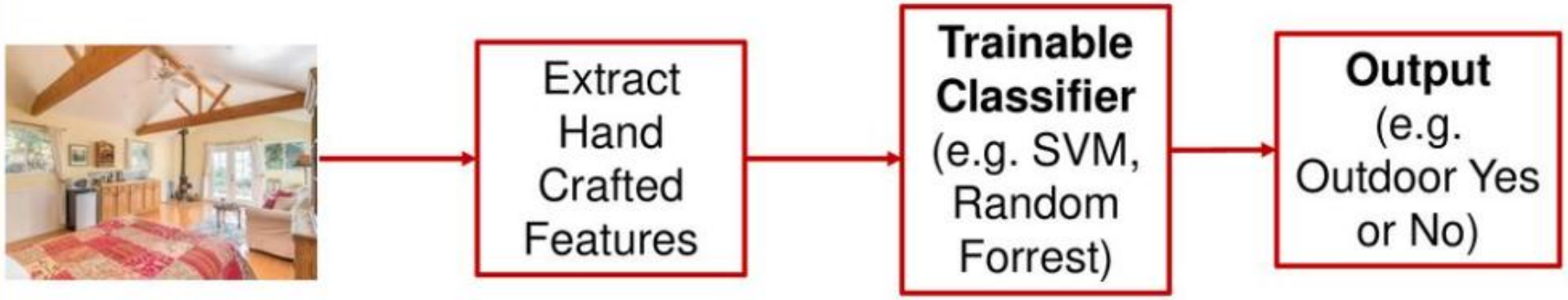
Output

Machine Learning vs. Deep Learning



Machine Learning vs. Deep Learning

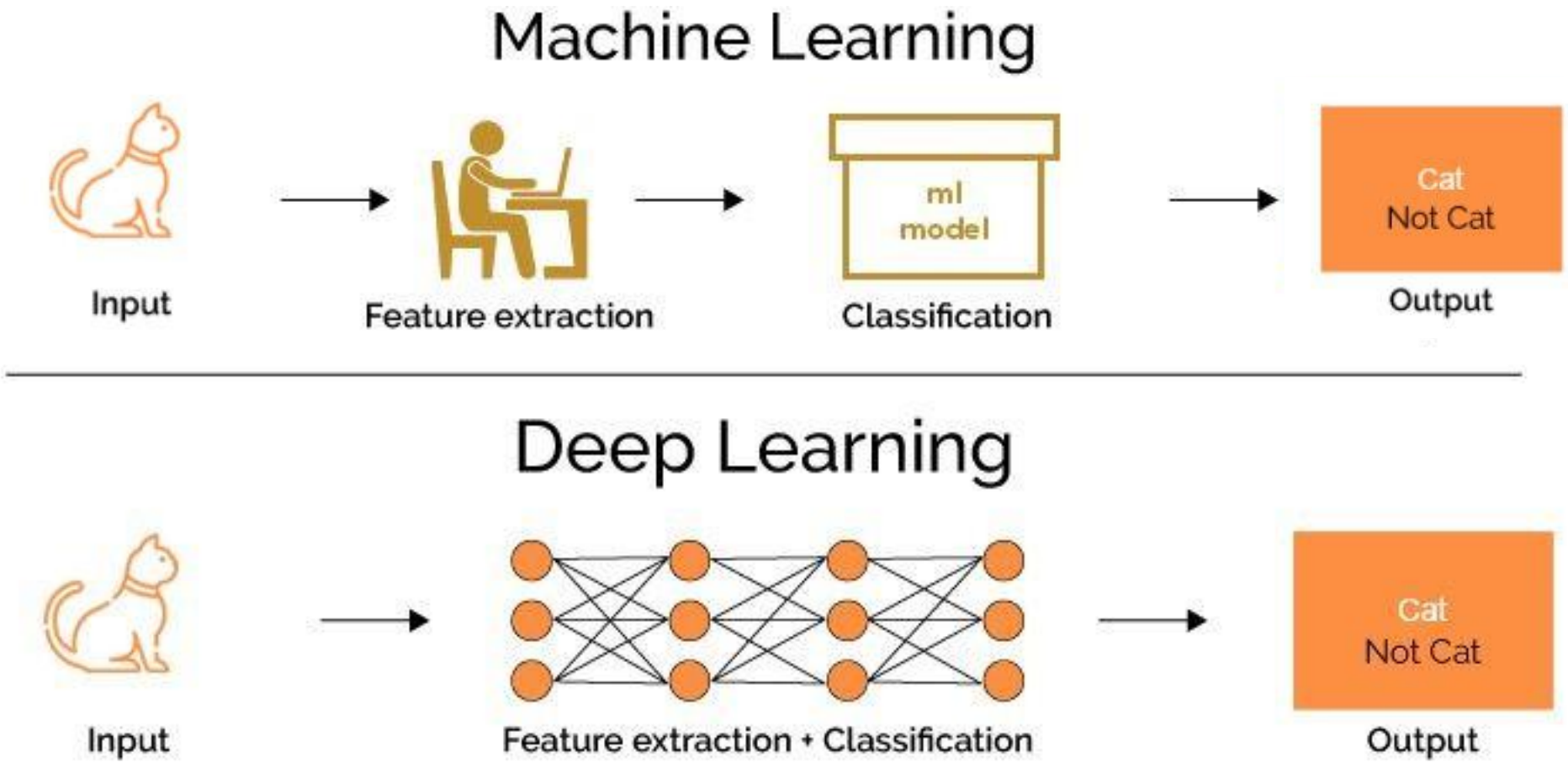
- Traditional pattern recognition models work with hand crafted features and relatively simple trainable classifiers.



Limitations

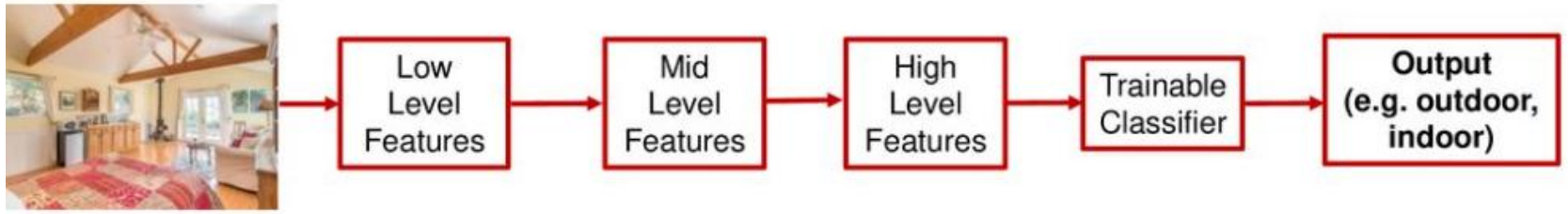
- Very tedious and costly to develop hand crafted features.
- The hand-crafted features are usually highly dependents on one application.

Machine Learning vs. Deep Learning



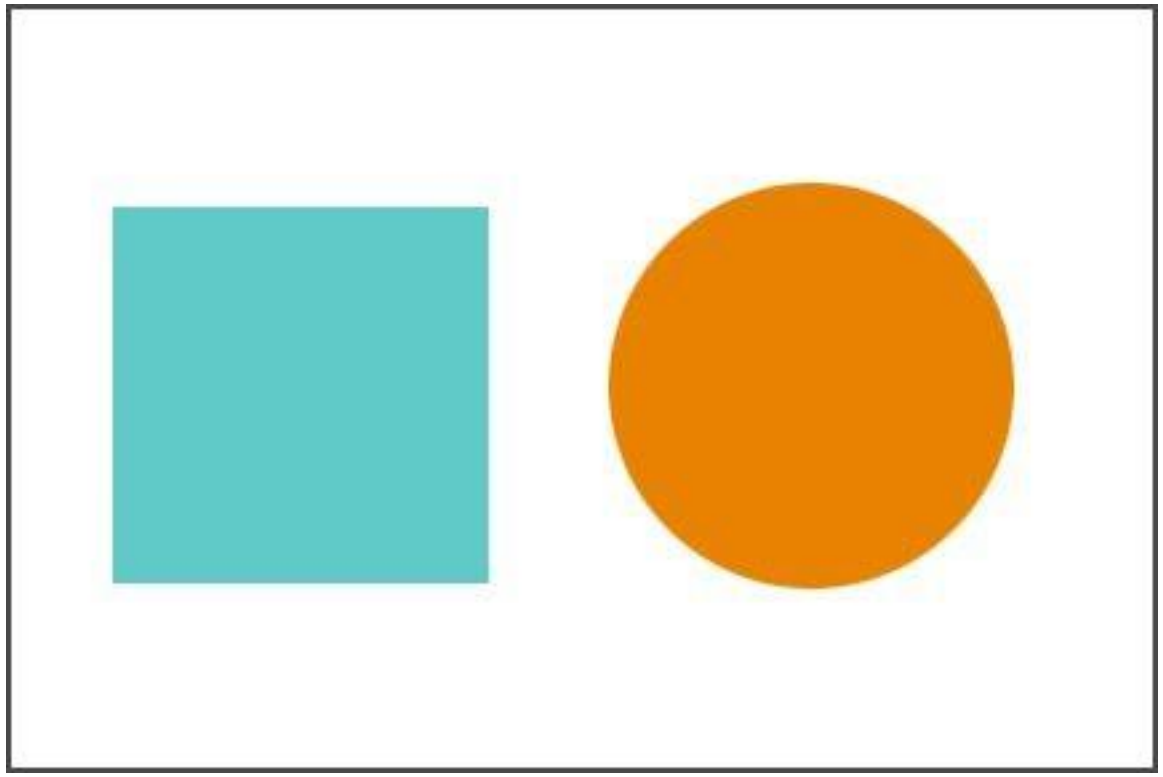
Machine Learning vs. Deep Learning

- Deep learning has an **inbuilt automatic multi stage feature learning process** that learns rich hierarchical representations (i.e. features).

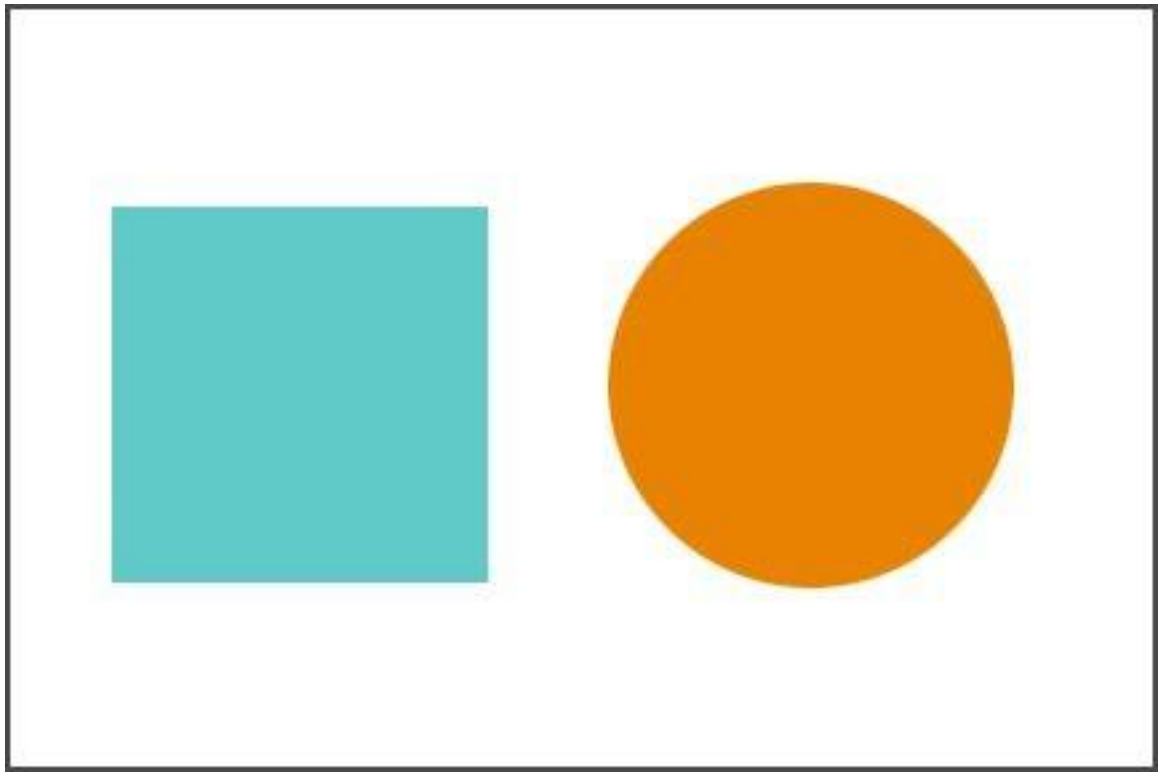




Represent the world as nested hierarchy of concepts



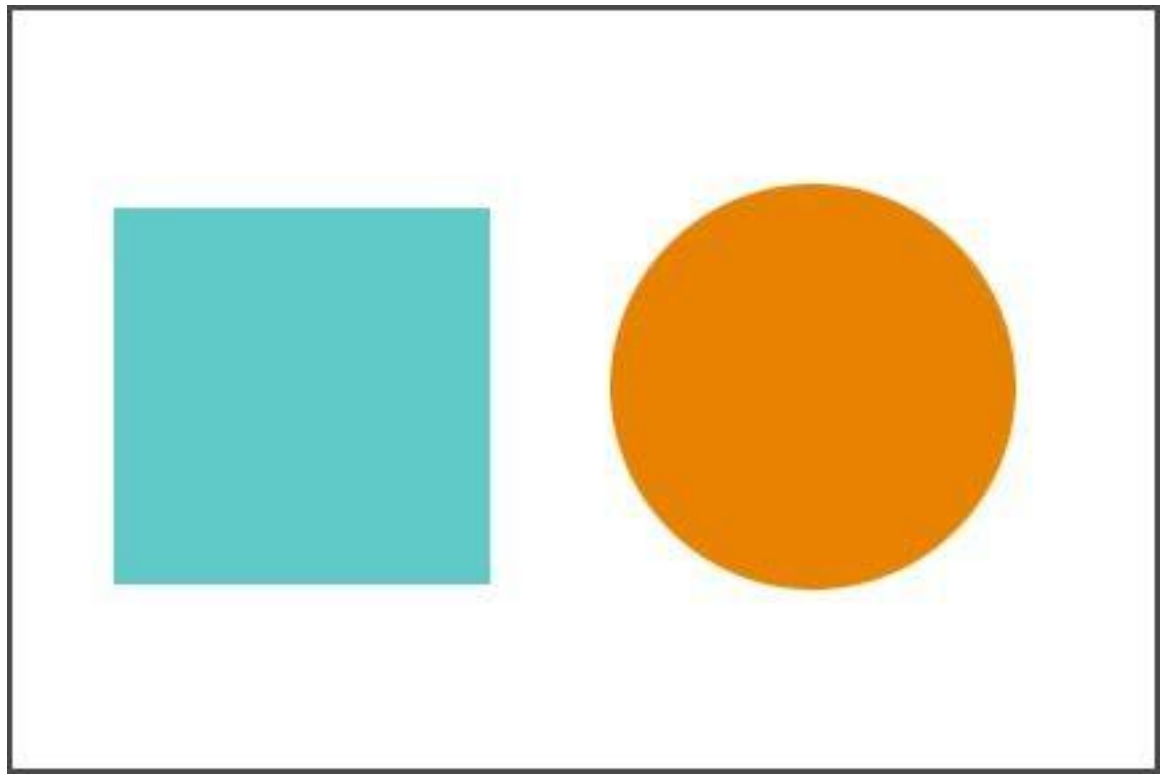
Represent the world as nested hierarchy of concepts



1. Four Lines

- Deep learning models learn representations hierarchically. For example, when learning to recognize a square:
- **First Level (Primitive Concepts):** The model learns very basic elements like individual lines.

Represent the world as nested hierarchy of concepts

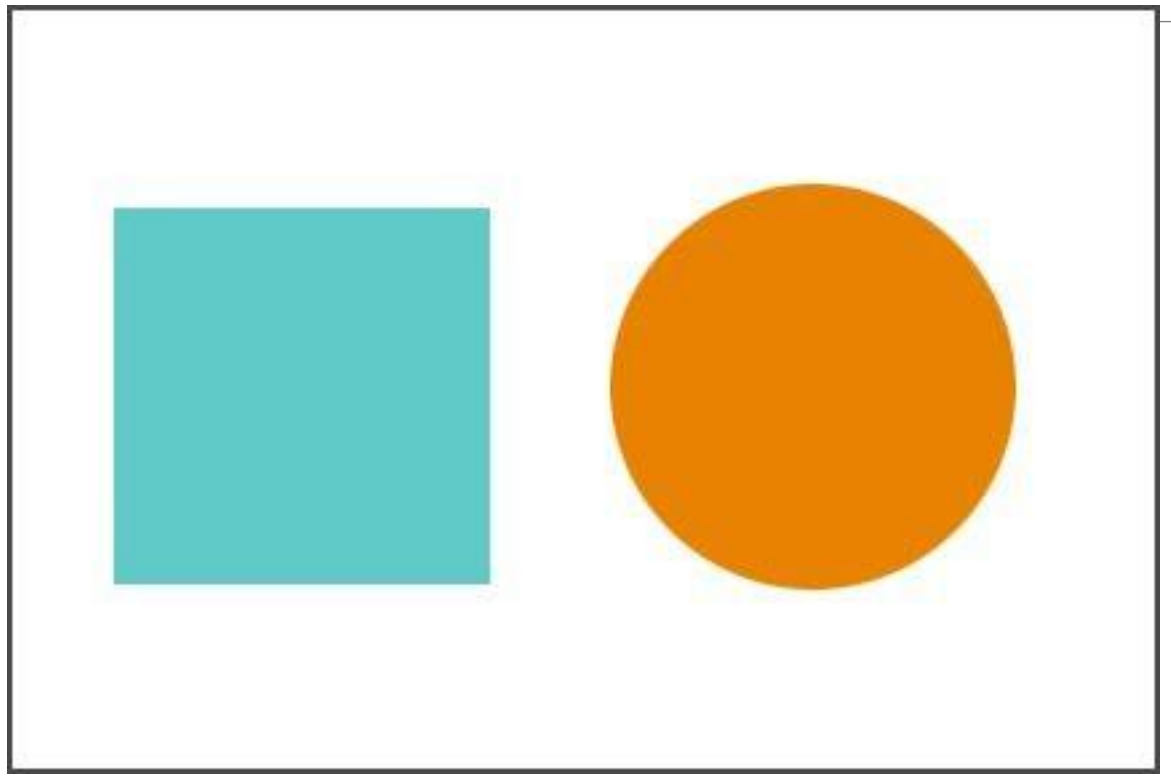


1. Four Lines
2. They are connected, Closed,

   perpendicular to each other

•**Second Level (Simple Relationships):** It learns that these lines are connected, form a closed loop, and are perpendicular to each other.

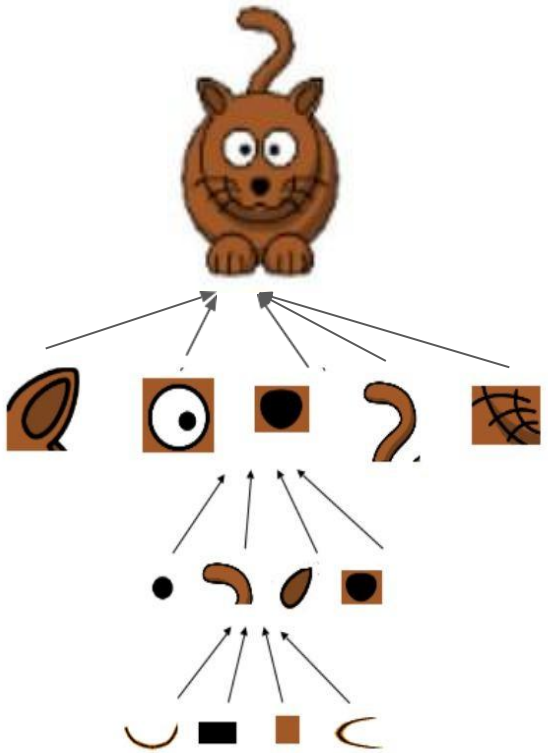
Represent the world as nested hierarchy of concepts



1. Four Lines
2. They are connected, Closed, perpendicular to each other
3. All lines are equal in length

• **Third Level (Complex Patterns):** The model learns that all the lines are equal in length, which collectively forms the concept of a square.

What is Deep Learning?



- Identifies what are the edges
- Builds on this hierarchically to find combination of shapes and edges

Layer Type	Features Learned	Examples	Purpose
Low-Level Features	Basic geometric shapes and patterns	Edges, lines, corners, simple textures, curves	Detects fundamental visual elements; serves as building blocks for more complex patterns.
Mid-Level Features	Combinations of basic shapes forming more complex structures	Parts of objects (e.g., eyes, ears, nose, mouth)	Identifies and recognizes object parts; helps in detecting patterns that signify object components.
High-Level Features	Abstract and holistic representation of complete objects	Entire objects (e.g., a complete cat, face, car)	Recognizes and classifies complete objects by combining all the lower and mid-level features.

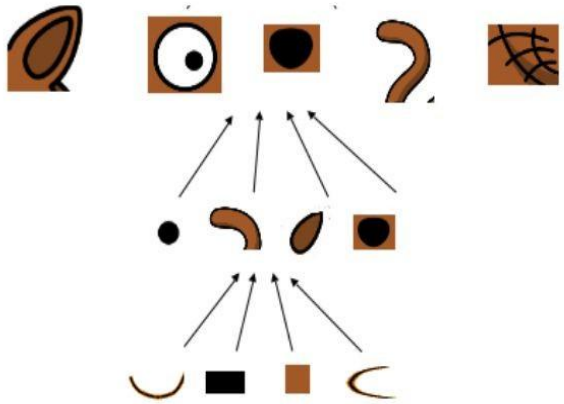
What is Deep Learning?





1. Identifies what are the edges

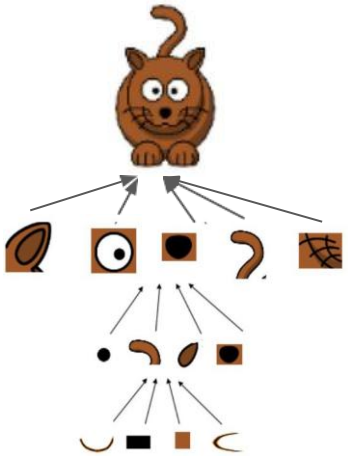
2. Builds on this hierarchically to find combination of shapes and edges



What is Deep Learning?



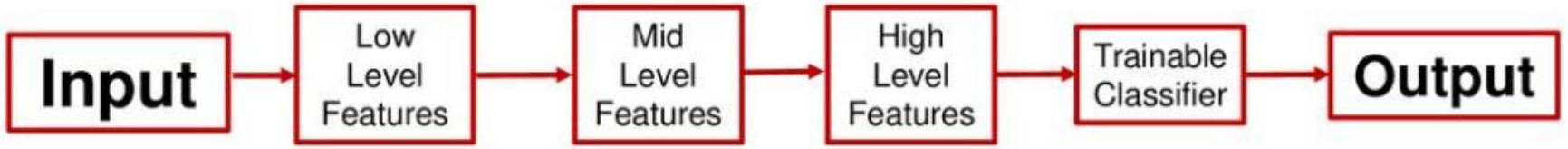
1. Identifies what are the edges



2. Builds on this hierarchically to find combination of shapes and edges



3. Identifies which of these features are responsible for Cat Vs. Dog classification



- Image

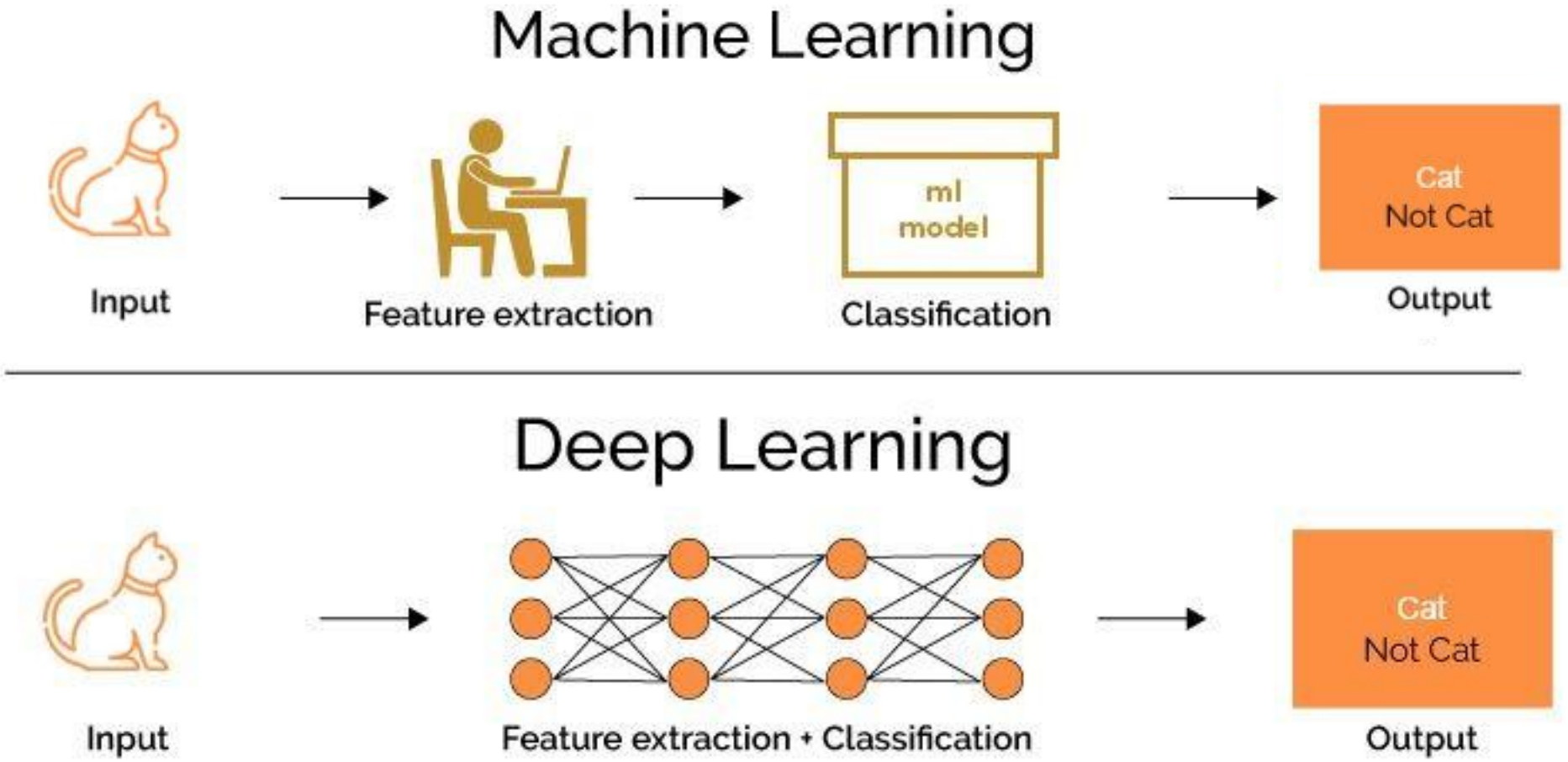
  Pixel → Edge → Texture → Motif → Part → Object

- Text

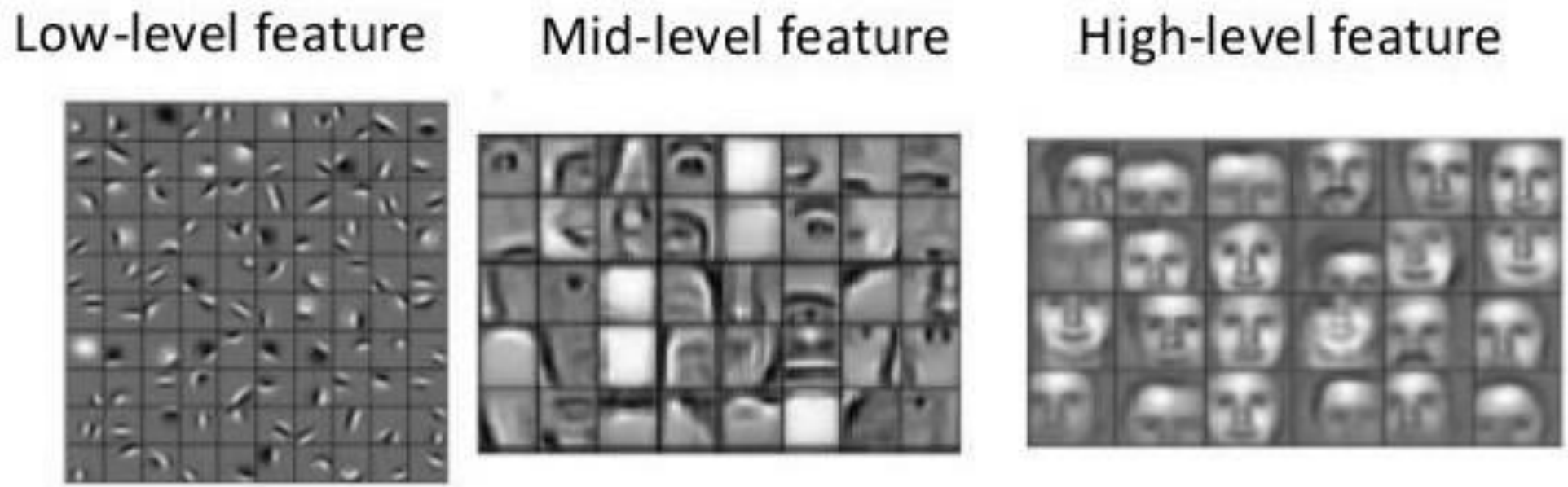
  Character → Word → Word-group → Clause → Sentence → Story

- Each module in Deep Learning transforms its input representation into a higher-level one, in a way similar to human cortex.

Automatic Feature Extraction

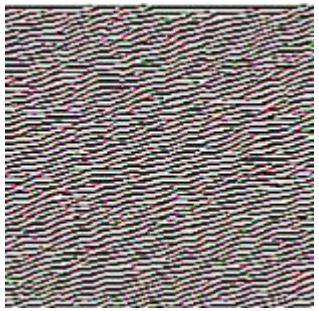


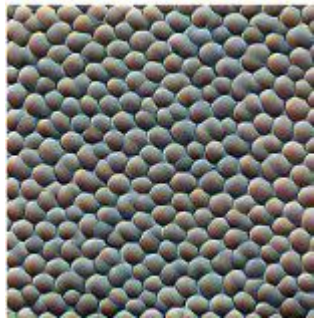
Automatic Feature Extraction



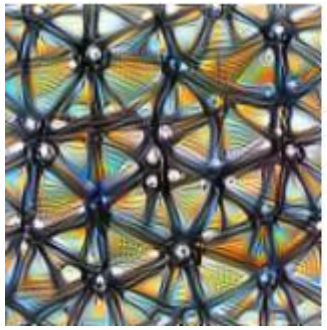
Source : Deep Learning in a Nutshell: Core Concepts, Nvidia
https://devblogs.nvidia.com/parallelforall/deep-learning-nutshell-core-concepts/

Data Type	Input	Low-Level Features	Mid-Level Features	High-Level Features	Trainable Classifier	Output
Image	Raw Image	Pixels, Edges	Textures, Motifs	Parts, Objects	Image Classifier	Object Recognition
Text	Raw Text	Characters, Words	Word Groups, Clauses	Sentences, Stories	Language Model	Text Generation
Audio	Raw Audio Waveform	Sound Wave Patterns, Amplitude	Phonemes, Timbre, Rhythm	Words, Melody, Genre	Speech Recognition, Genre Classifier	Recognized Speech, Classified Genre
Sensor	Raw Sensor Data	Measurements (Temp, Pressure)	Trends, Anomalies	Behavioral Patterns	Predictive Maintenance	Alerts, Predictions
Financial	Raw Market Data	Price Movements, Volatility	Technical Indicators, Patterns	Market Sentiment, Trend Analysis	Stock Price Predictor	Buy/Sell Recommendations
Medical	Raw Medical Images	Pixels, Edges, Intensities	Tissue Structures, Textures	Organ, Lesion Detection	Disease Detection	Diagnosis, Treatment Recommendations
Social Media	Raw Text, Images, Videos	Words, Hashtags, Pixels	Sentiments, Topics	User Patterns, Influencer Identification	Sentiment Analysis	Advertisements, Content Suggestions
Genomic	Genetic Sequences	Nucleotide Bases (A, T, C, G)	Codons, Gene Sequences	Gene Functions, Variations	Disease Risk Predictor	Personalized Medicine















INPUT IMAGE



OUTPUT CLASSES

- Labrador Retriever
- Golden Retriever
- Tennis Ball
- Rhodesian Ridge...
- Appenzeller

TOP CHANNELS SUPPORTING LABRADOR RETRIEVER

MIXED3B







MIXED4A

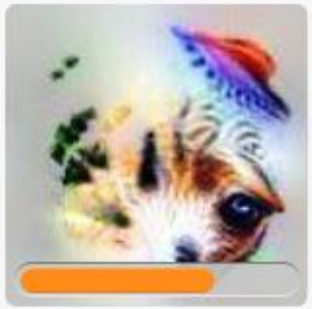






MIXED4B







MIXED4C







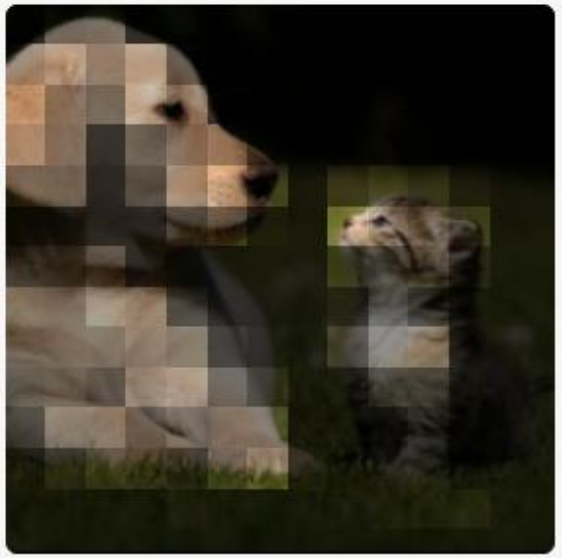
MIXED4D







INPUT IMAGE



OUTPUT CLASSES

Vizsla

Brittany Spaniel

Chesapeake Bay ...

Lakeland Terrier

Labrador Retriever

TOP CHANNELS SUPPORTING LABRADOR RETRIEVER

MIXED3B | MIXED4A | MIXED4B | MIXED4C | MIXED4D



https://distill.pub/2018/building-blocks/

INPUT IMAGE



OUTPUT CLASSES

- Lemon
- Vase
- Pitcher
- Water Jug
- Orange

TOP CHANNELS SUPPORTING LEMON

MIXED3B







MIXED4A







MIXED4B







MIXED4C







MIXED4D





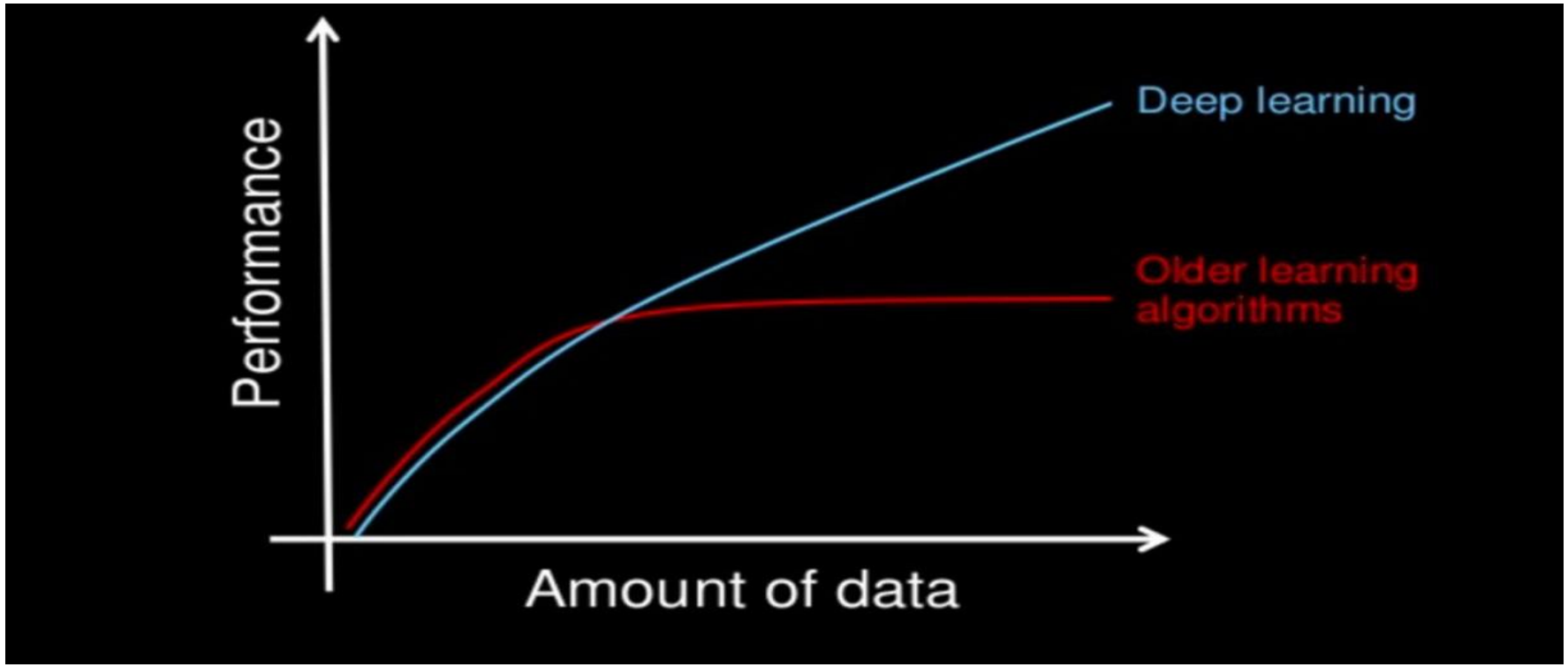


What is Deep Learning?

*"Deep learning is a **particular kind of machine learning** that achieves great power and flexibility by **learning to represent** the world as **nested hierarchy of concepts**, with each concept defined in relation to simpler concepts, and more abstract representations computed in terms of less abstract ones"*


Performs well with high amount of Data

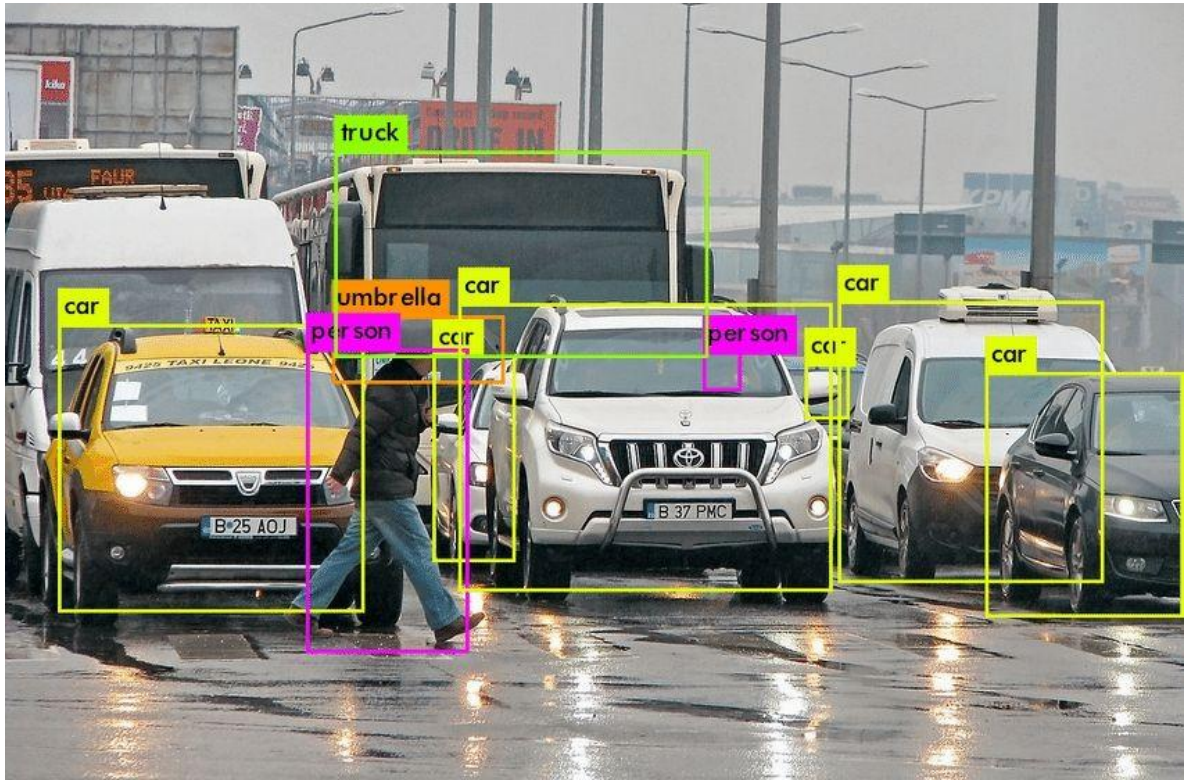
Performs well with high amount of Data



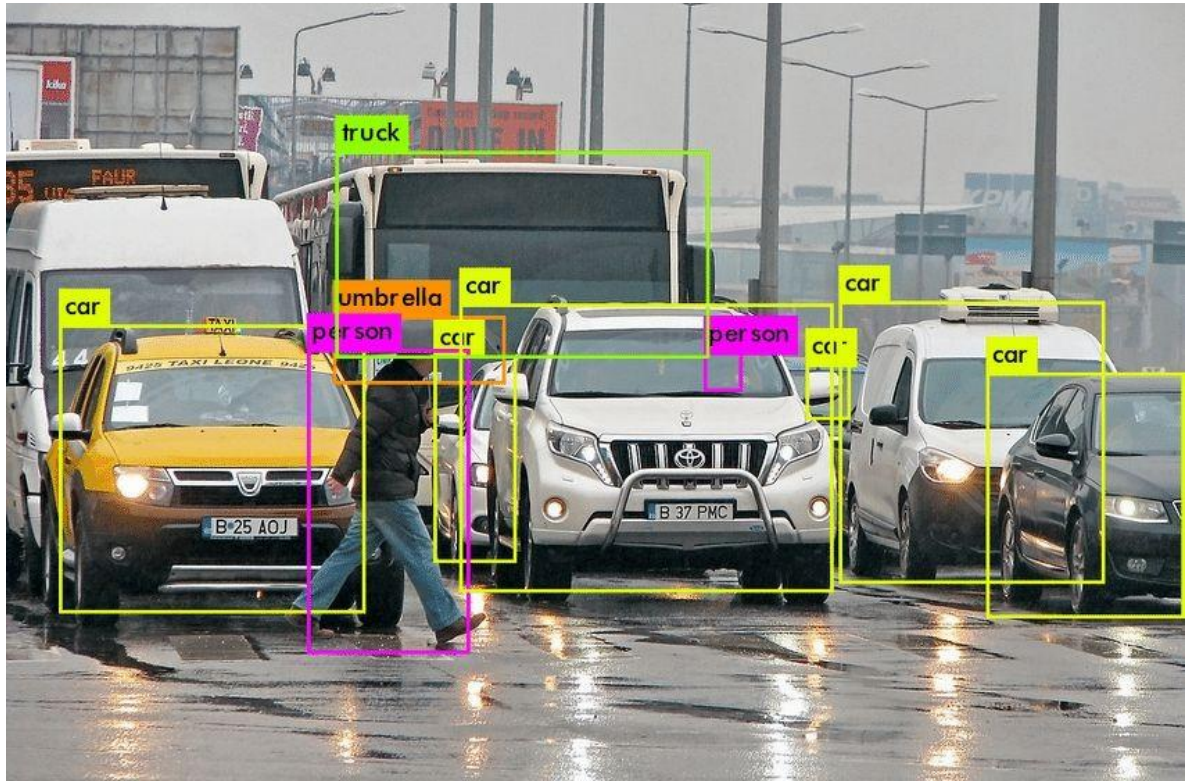
Deep Learning algorithms require more computation power



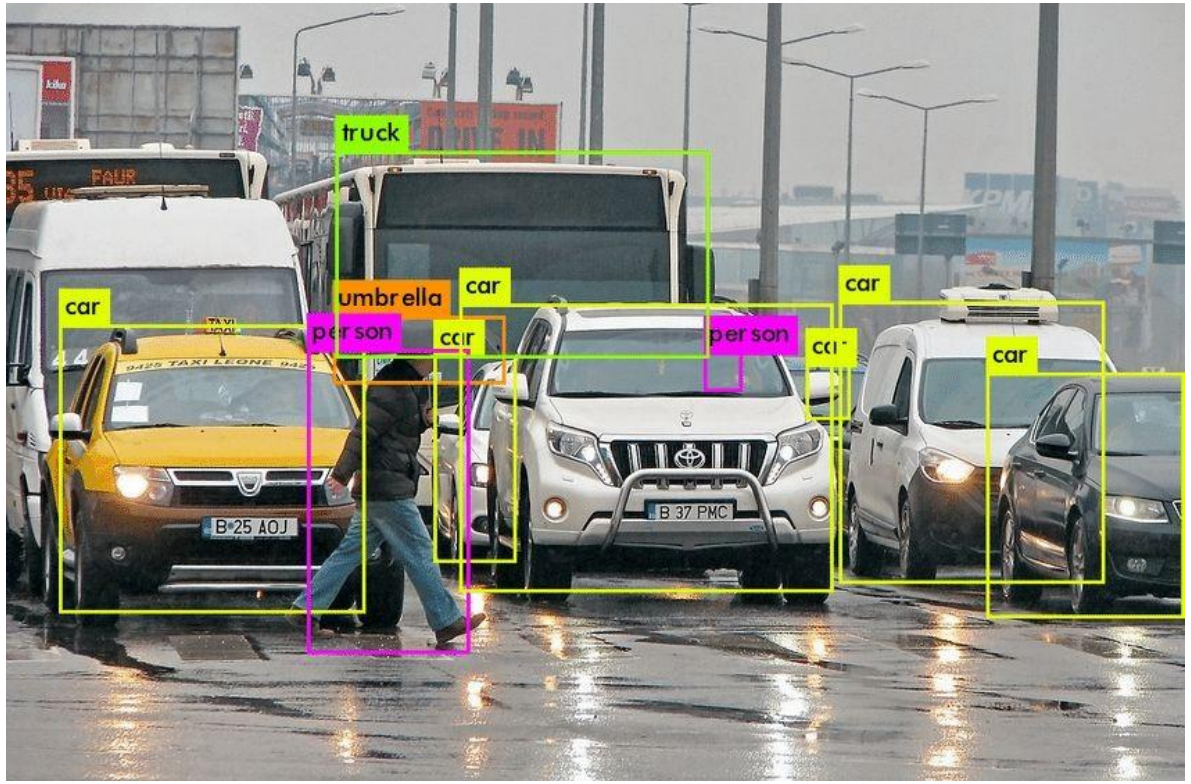
Deep Learning helps to solve problems End -to- End



Deep Learning helps to solve problems End -to- End

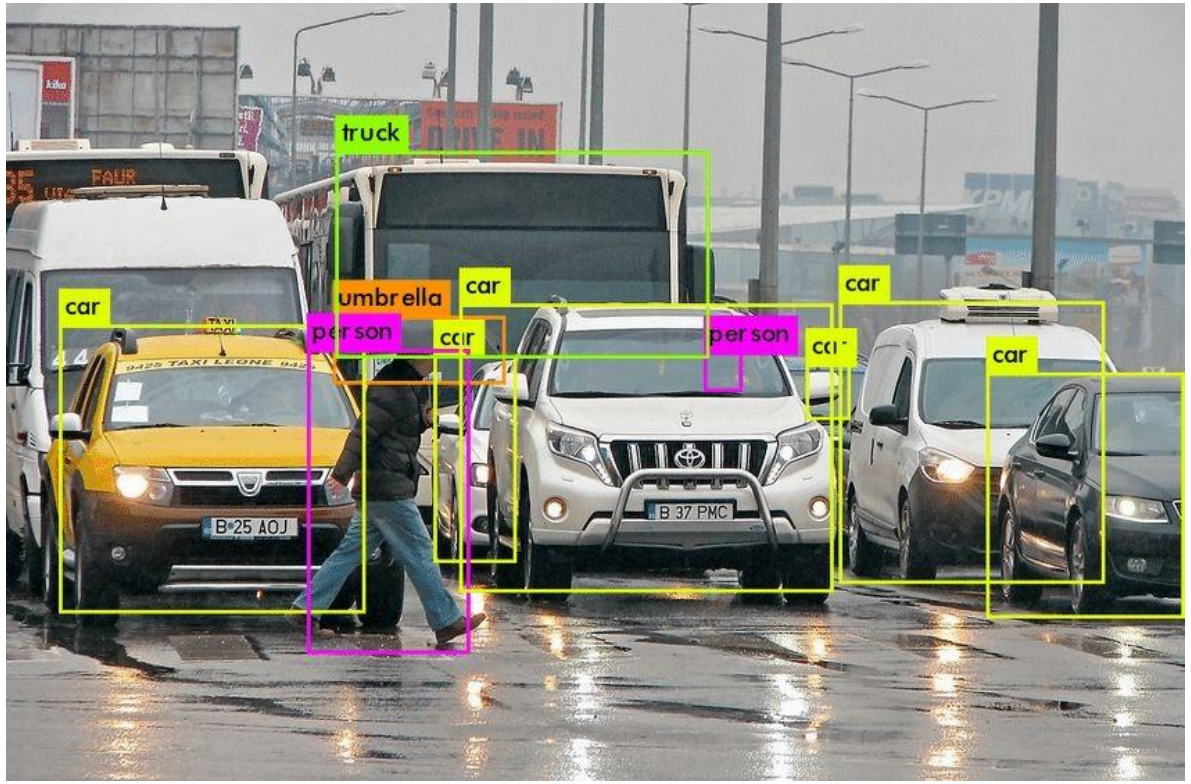


Deep Learning helps to solve problems End-to-End



1. Object Detection

Deep Learning helps to solve problems End -to- End

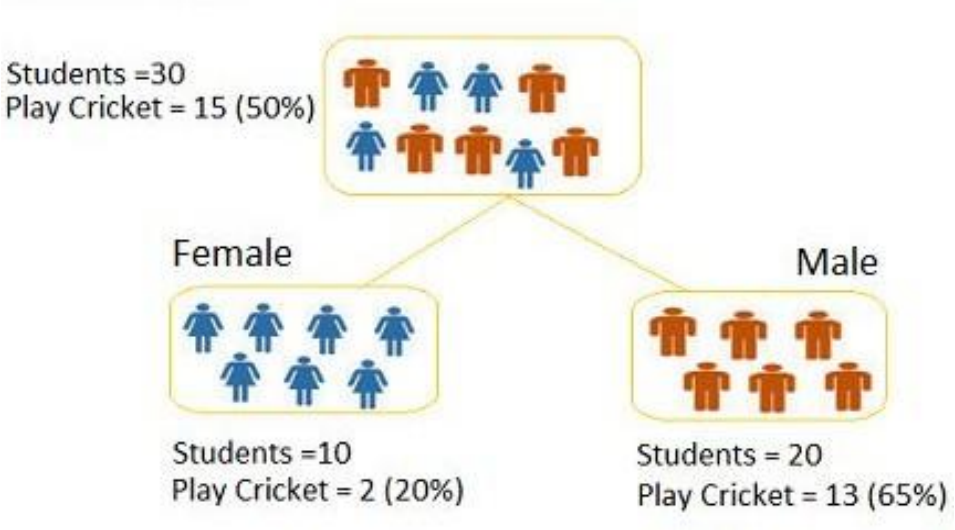


1. Object Detection

2. Object Recognition

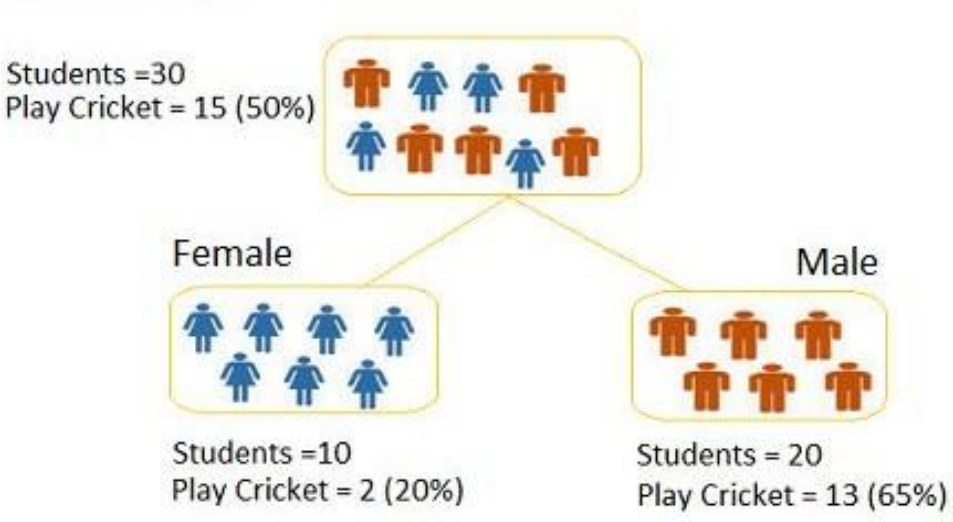
Deep Learning is less interpretable

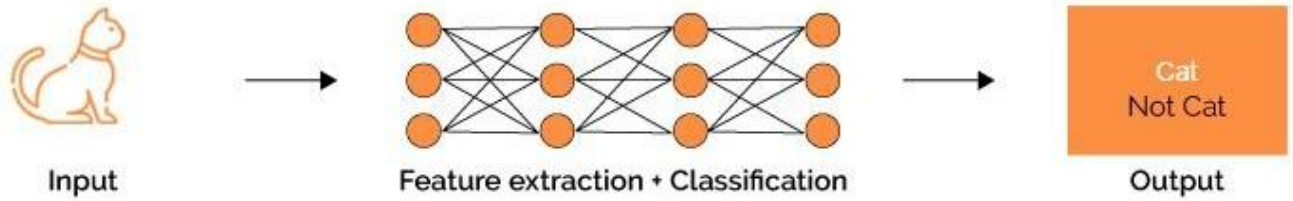
Deep Learning is less interpretable



The tree starts with 30 students, out of which 15 (50%) play cricket.
It then splits the students into two groups: "Female" and "Male.
"For the "Female" group: Out of 10 students, only 2 (20%) play cricket.
For the "Male" group: Out of 20 students, 13 (65%) play cricket.
The decision tree clearly shows the decision-making process, where you can see how each input (e.g., gender) affects the output (e.g., the likelihood of playing cricket).
The logic is explicit and easy to follow.

Deep Learning is less interpretable





The decision tree clearly shows the decision-making process, where you can see how each input (e.g., gender) affects the output (e.g., the likelihood of playing cricket). The logic is explicit and easy to follow.

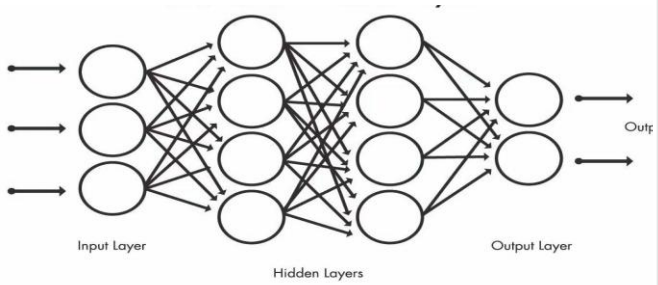
Unlike the decision tree, the deep learning model does not provide clear insight into how it reaches its conclusions. The decision-making process involves many hidden layers and a complex combination of weights and activations, which are not easy for humans to interpret.

Machine Learning Vs. Deep Learning

Automatic Feature Extraction	Performs well on More Data
Less Interpretable	Requires More Computation Power

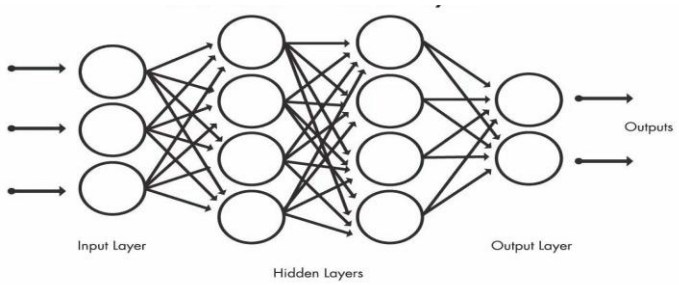
Why Deep Learning has become so popular among data scientists?

Why has Deep Learning become so popular?



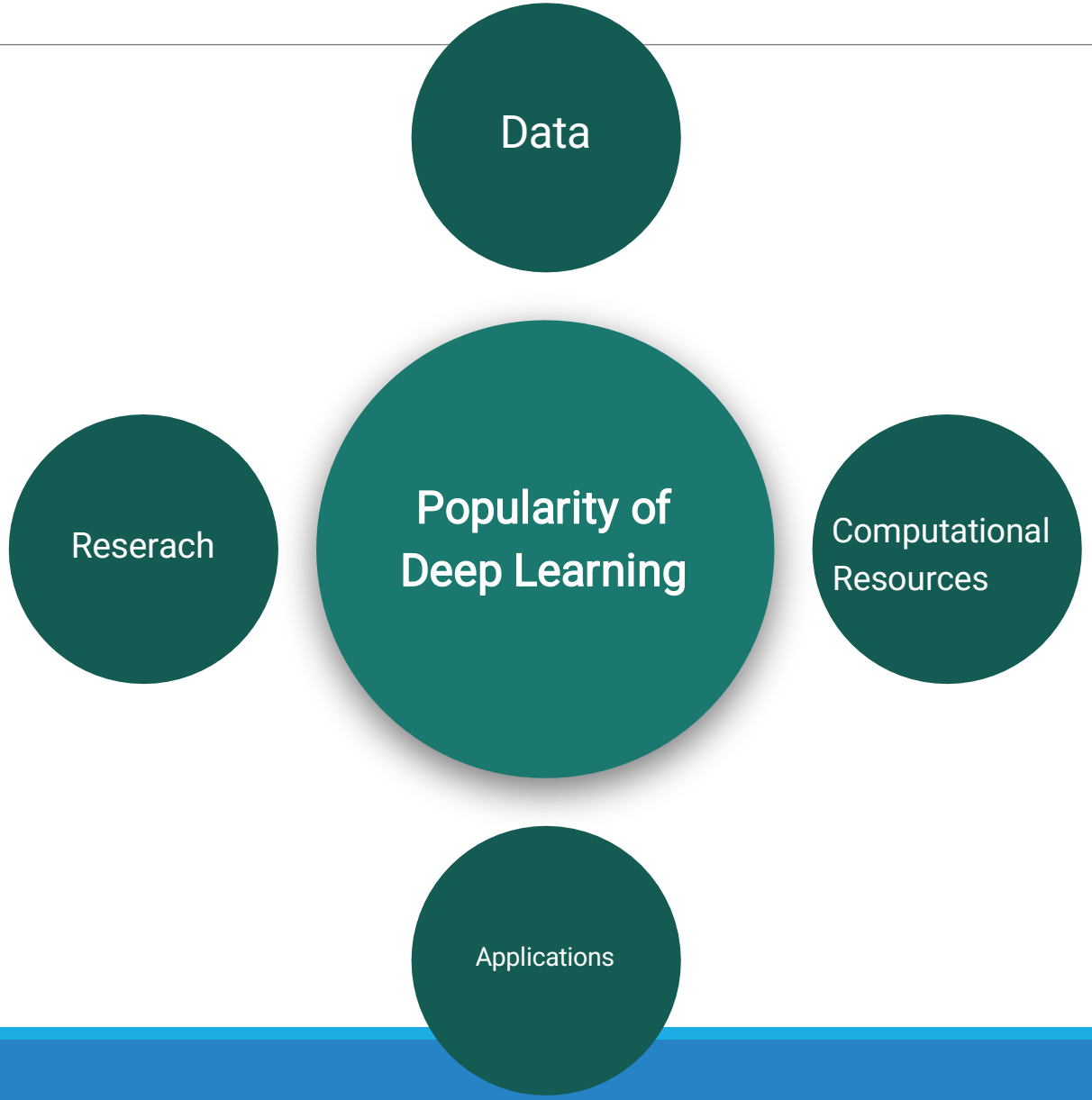
Date	Event	Description
1943	**McCulloch-Pitts Neuron Model**	Warren McCulloch and Walter Pitts proposed a simplified model of a neuron, laying the foundation for artificial neural networks.
1958	**Perceptron Introduced**	Frank Rosenblatt developed the perceptron, an early type of neural network capable of binary classification.
1969	**Limitations of Perceptrons**	Marvin Minsky and Seymour Papert published "Perceptrons," highlighting limitations of single-layer perceptrons, contributing to a decline in interest.
1970s-1980s	**First AI Winter**	A period of reduced funding and interest in AI research, partly due to the limitations identified in neural networks and unrealistic expectations.

Why has Deep Learning become so popular?



1986	**Backpropagation Popularized**	Geoffrey Hinton, David Rumelhart, and Ronald J. Williams published a paper on the backpropagation algorithm, revitalizing neural network research.
1990s	**Resurgence of Neural Networks**	Increased computational power and the advent of new algorithms led to renewed interest in neural networks.
2006	**Deep Learning Revolution Begins**	Geoffrey Hinton and collaborators introduced deep belief networks, marking the start of modern deep learning.
2010	**Rise of GPU Computing**	Graphics Processing Units (GPUs) were increasingly used to accelerate neural network training.
2012	**ImageNet Competition Breakthrough**	Geoffrey Hinton's team, including Alex Krizhevsky and Ilya Sutskever, won the ImageNet competition using a deep convolutional neural network (AlexNet), achieving a significant leap in accuracy.

Reasons behind this attention

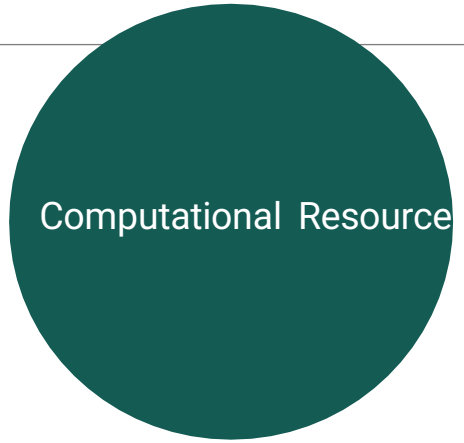


Reasons behind this attention

Data

Factor	Description
Data	- **Abundance of Data**: The massive amount of data generated daily from various sources provides deep learning models with the data needed to learn complex patterns. - **Improved Data Quality**: High-quality labeled datasets enable effective supervised learning, improving the performance of deep learning models. - **Diverse Data Types**: Deep learning can handle various data types (text, images, audio, video), making it applicable across multiple domains.

Reasons behind this attention



Computational Resources	- **Advancements in Hardware**: GPUs and TPUs enable efficient parallel computations required for training large neural networks. - **Cloud Computing**: Provides scalable and affordable access to vast computational resources, reducing the need for costly hardware investments. - **Specialized AI Chips**: AI chips, such as those developed by NVIDIA and Google, enhance computational power and reduce energy consumption.

Reasons behind this attention



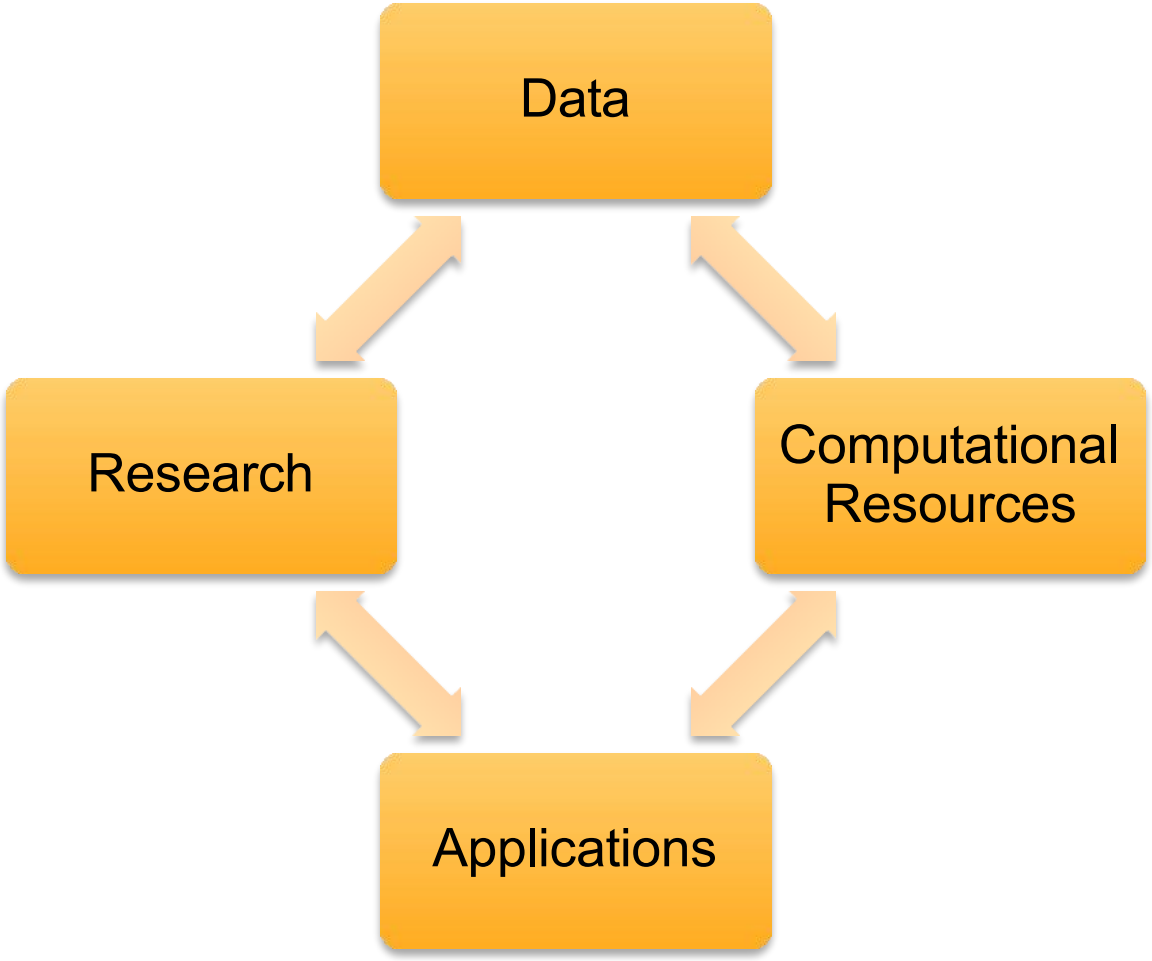
Applications	- **Wide Range of Applications**: Deep learning is used in many fields, including healthcare, finance, entertainment, and transportation. - **Breakthroughs in Key Areas**: Success in fields like computer vision, NLP, and autonomous systems has showcased deep learning's potential. - **Enhanced User Experiences**: Personalized and seamless experiences in daily life through applications like voice assistants and recommendation systems.

Reasons behind this attention

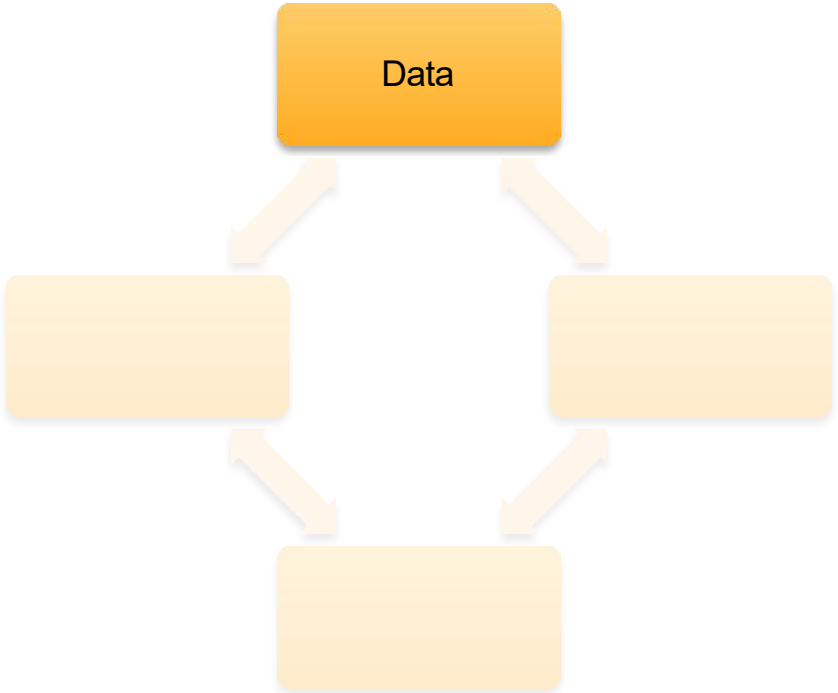


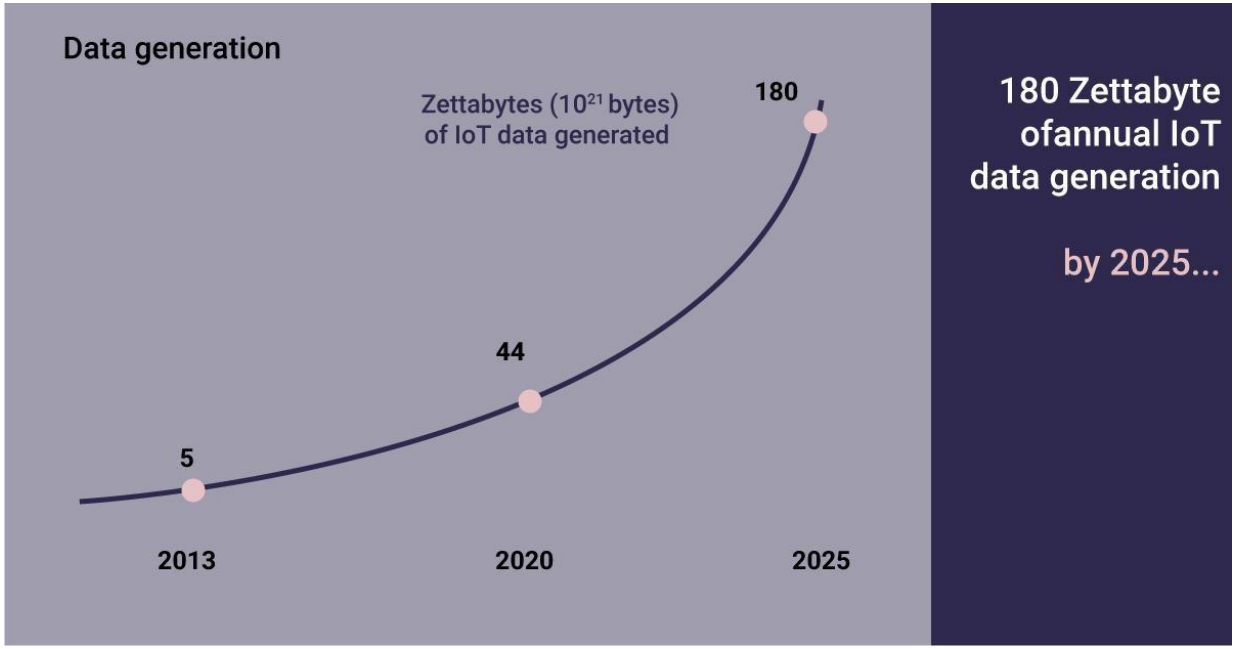
Research	- **Theoretical and Algorithmic Advancements**: Improved architectures (e.g., CNNs, RNNs, LSTMs, transformers) and new approaches (e.g., GANs, transfer learning) have expanded the capabilities of deep learning. - **Open Source and Collaboration**: Open-source frameworks (e.g., TensorFlow, PyTorch, Keras) and collaborative research have accelerated innovation and development. - **Increasing Academic and Industry Focus**: Growing number of research papers, conferences, and workshops dedicated to deep learning topics, fostering a dynamic and innovative ecosystem.

Reasons behind this attention



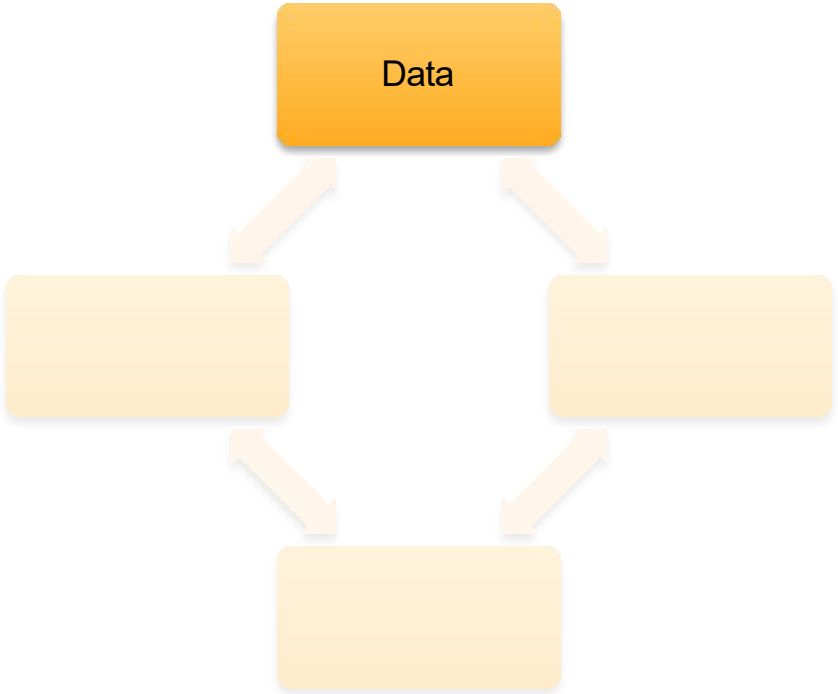
Exponential Data Growth since 2013

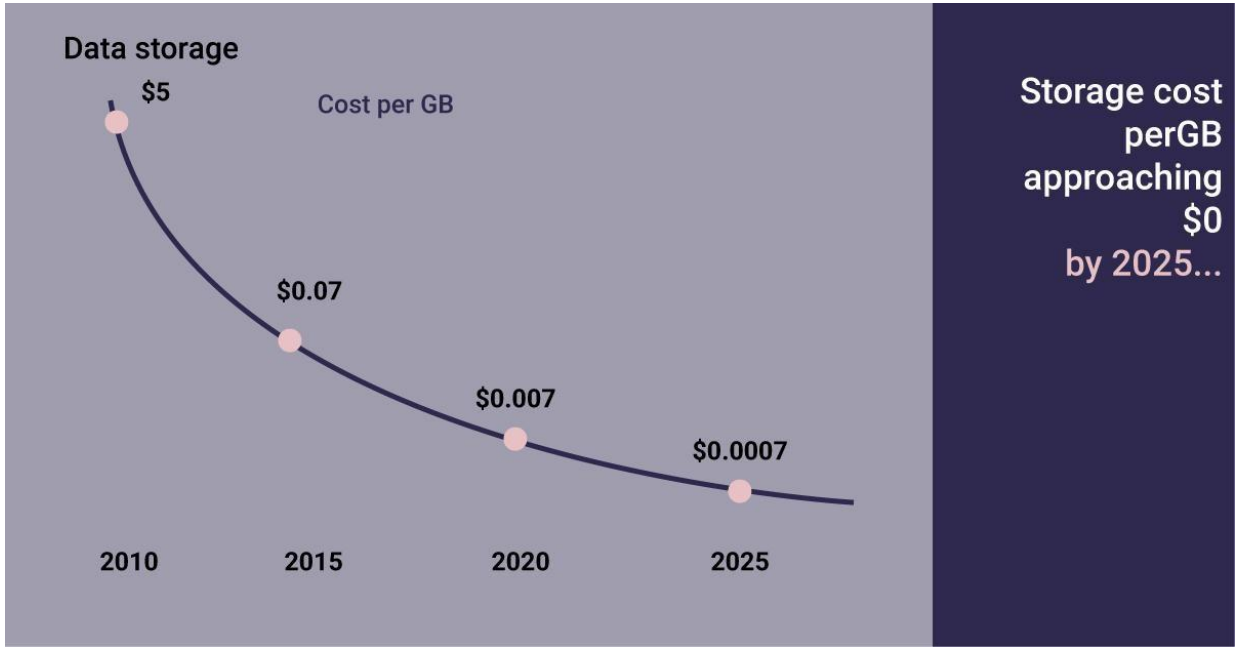




Source: https://fortune.com/2018/09/24/business-strategy-technology-mckinsey/

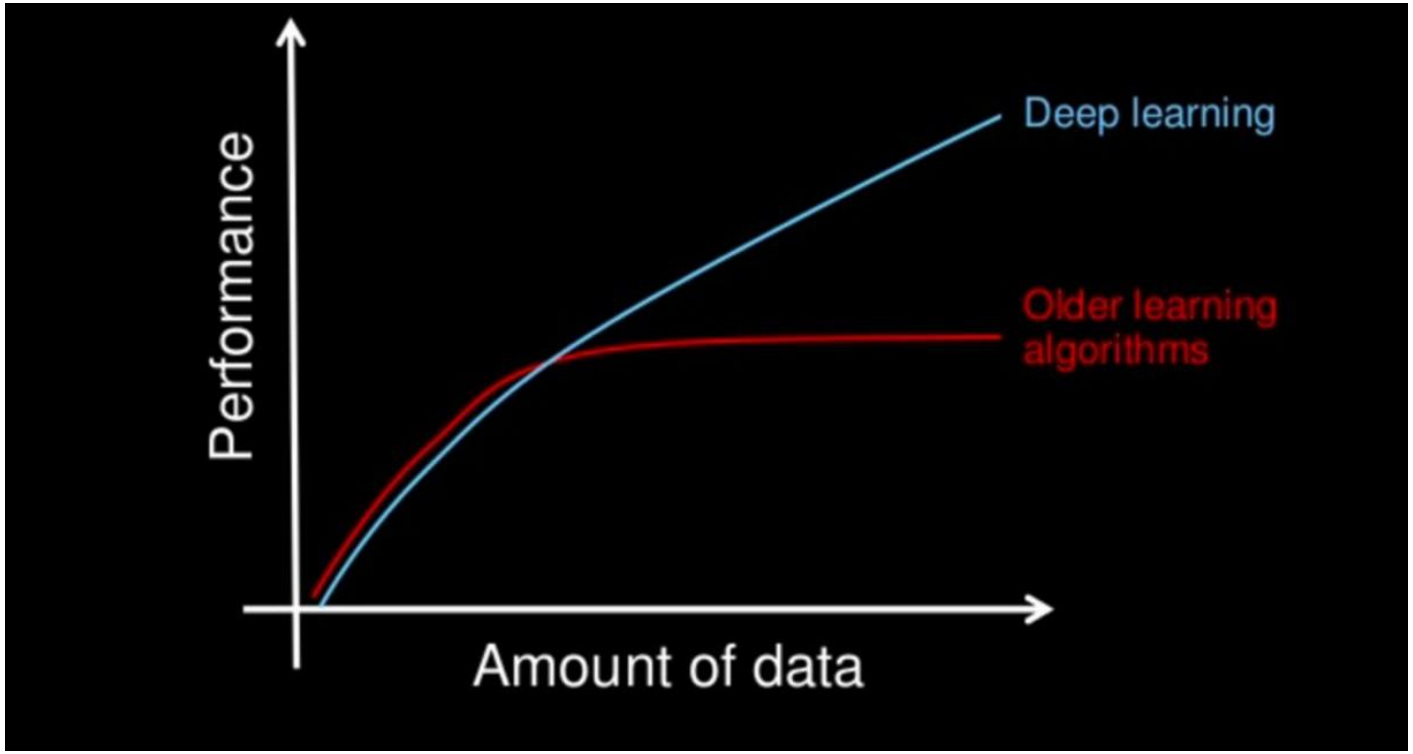
Reduction in Data Storage cost



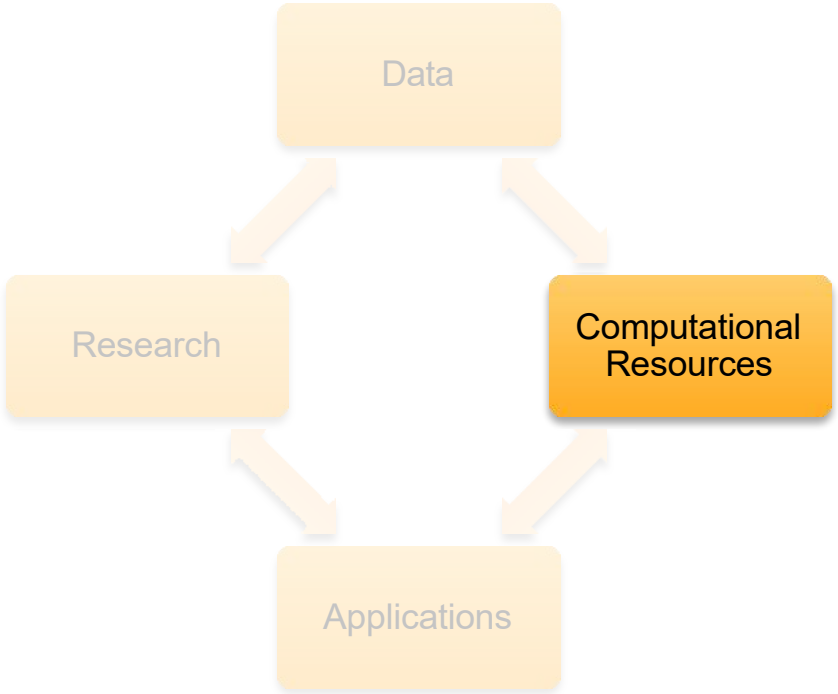


Source: https://fortune.com/2018/09/24/business-strategy-technology-mckinsey/

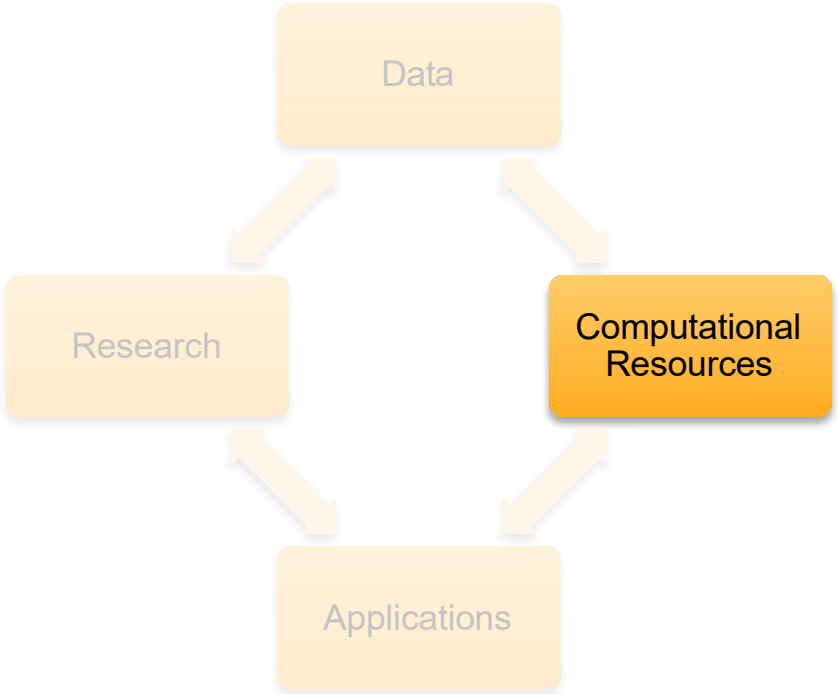
Performs well with high amount of Data

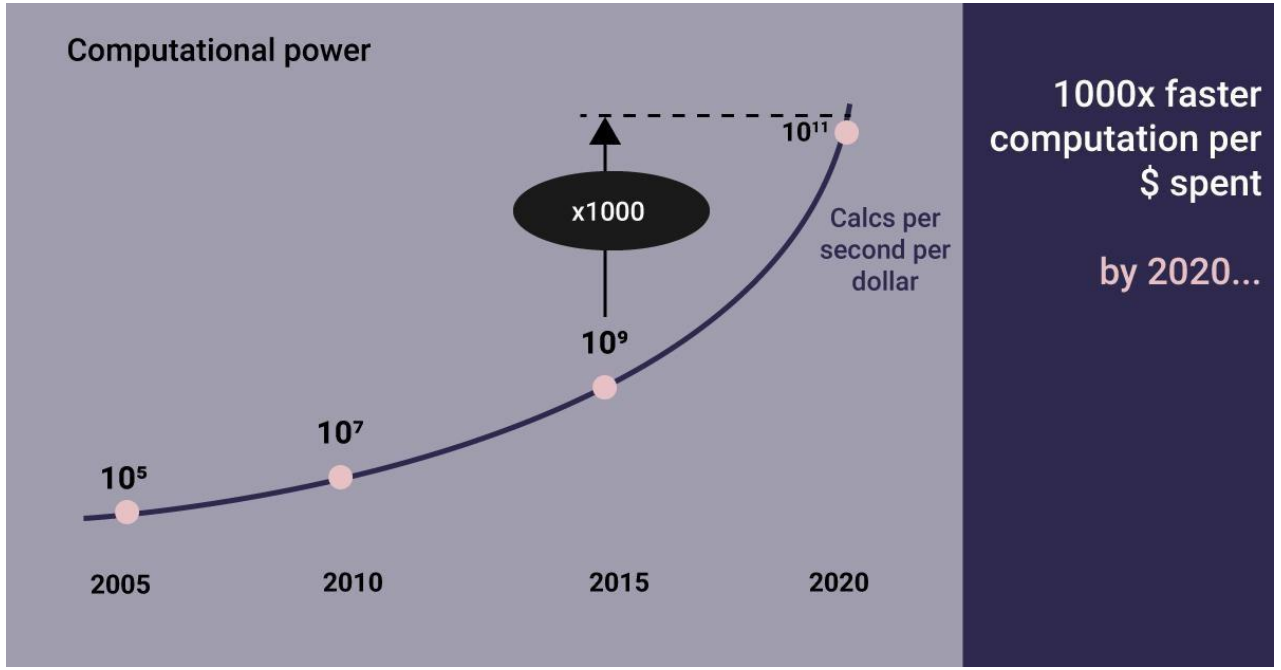


The Computational Power increased by 100,000 times since 2000



The Computational Power increased by 1000 times since 2015





Source: https://fortune.com/2018/09/24/business-strategy-technology-mckinsey/

CPU vs GPU vs TPU



CPU



GPU



TPU

Cloud GPU/ TPU Alternatives for Deep Learning

- Google Colaboratory (Both GPU and TPU)
- Kaggle Kernels
- Paperspace
- Vast.AI
- Oracle Cloud
- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Deep Learning Framework





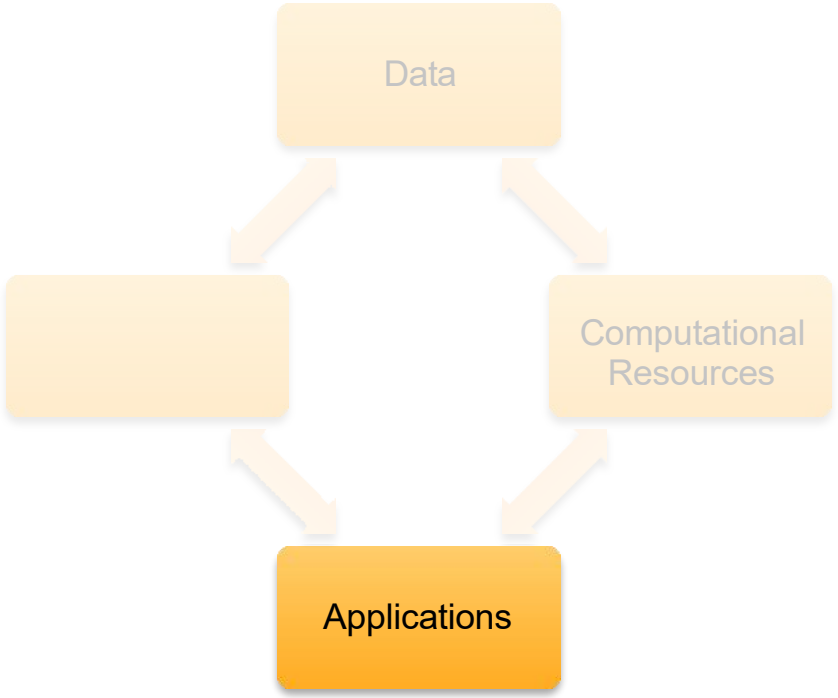




What kind of Hardware is needed for Deep Learning?

1. **Graphics Processing Unit (GPU)**
    a. NVIDIA GTX 1060, GTX 1070, GTX 1080 and above
2. **RAM**
    a. 8 GB and above
3. **HDD/SSD**
    a. 1TB HDD (datasets) and good to have 256GB SSD (faster preprocessing)
4. **CPU**
    a. i5 8th Gen and above

Availability of Data and Computational Resources leads to more Applications



Neural Network was there since 1960's but due to less availability of data and computer resources, we were not able to apply in industry

Applications of Deep Learning: Automobile



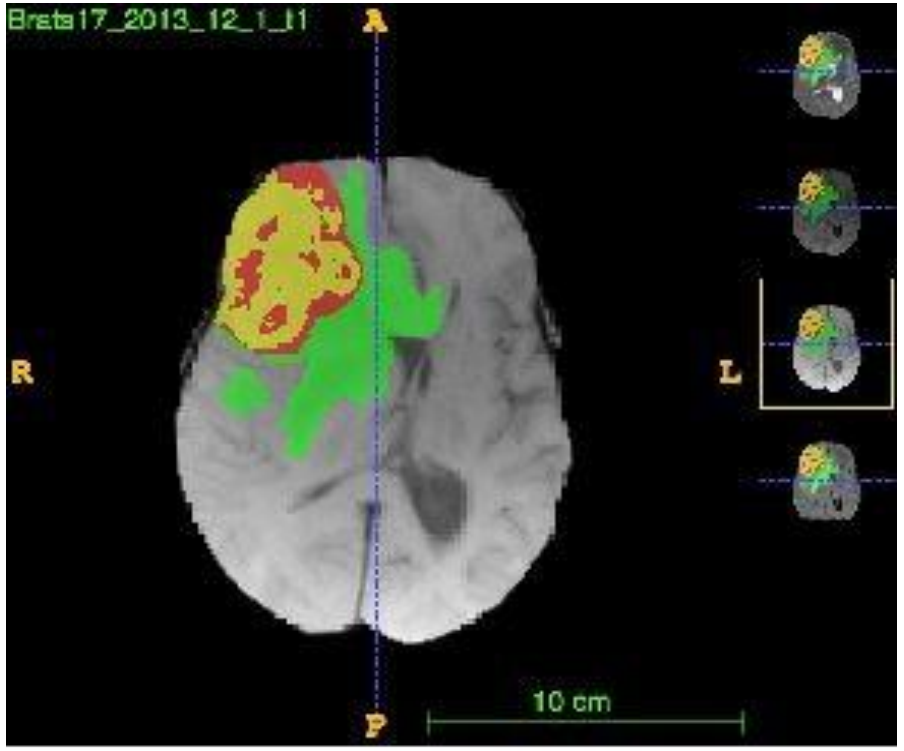
Deep learning has been instrumental in the development of self-driving cars. These vehicles rely on complex algorithms, advanced sensors, and deep neural networks to perceive their environment, make decisions, and navigate safely without human intervention.

Applications of Deep Learning: Automobile

Real-World Examples and Current Developments

1. **Waymo (Google's Self-Driving Car Project)**:
    1. Waymo, initially a Google project, is one of the leading companies in developing self-driving cars. It uses deep learning algorithms extensively for perception, decision-making, and path planning. Waymo's cars have driven millions of miles on public roads and in simulation, continually improving their safety and reliability.
2. **Tesla Autopilot**:
    1. Tesla's Autopilot and Full Self-Driving (FSD) systems rely on neural networks to interpret data from cameras and sensors, navigate, and control the vehicle in various driving conditions. Tesla's approach focuses on vision-based deep learning, using large amounts of data collected from its fleet of cars to train its models.
3. **NVIDIA Drive**:
    1. NVIDIA has developed the Drive platform, a complete hardware and software stack for autonomous vehicles. It uses deep learning to handle perception, mapping, and control tasks. The platform is designed to enable Level 4 and Level 5 autonomous driving, where the car can handle all driving tasks without human intervention.
4. **Uber Advanced Technologies Group (ATG)**:
    1. Uber ATG has worked on developing self-driving technologies that use deep learning for perception and decision-making. Although the project faced challenges, the research has contributed to advancements in autonomous vehicle technology.

Applications of Deep Learning: Healthcare



The image shows a brain scan where deep learning algorithms have been used to identify and highlight specific regions, such as a tumor or abnormal tissue. The different colors represent distinct areas of interest—such as lesions, edema (swelling), or healthy tissue—detected by the model.

Applications of Deep Learning: Healthcare

Real-World Examples of Deep Learning in Medical Imaging

•**Google Health's Deep Learning Models**: Google Health has developed deep learning models to analyze medical images, including retinal scans to detect diabetic retinopathy and lung scans for cancer detection. These models have shown high accuracy rates, sometimes surpassing human experts.

•**Enlitic and Zebra Medical Vision**: Companies like Enlitic and Zebra Medical Vision use AI and deep learning to detect diseases from X-rays, CT scans, and other types of medical imaging. Their tools help radiologists make more accurate and faster diagnoses.

•**DeepMind's AI for Eye Disease**: DeepMind (a subsidiary of Alphabet) collaborated with Moorfields Eye Hospital in London to develop an AI system capable of detecting over 50 different eye diseases from 3D OCT scans, potentially speeding up diagnosis and treatment.

Applications of Deep Learning: AI assistant

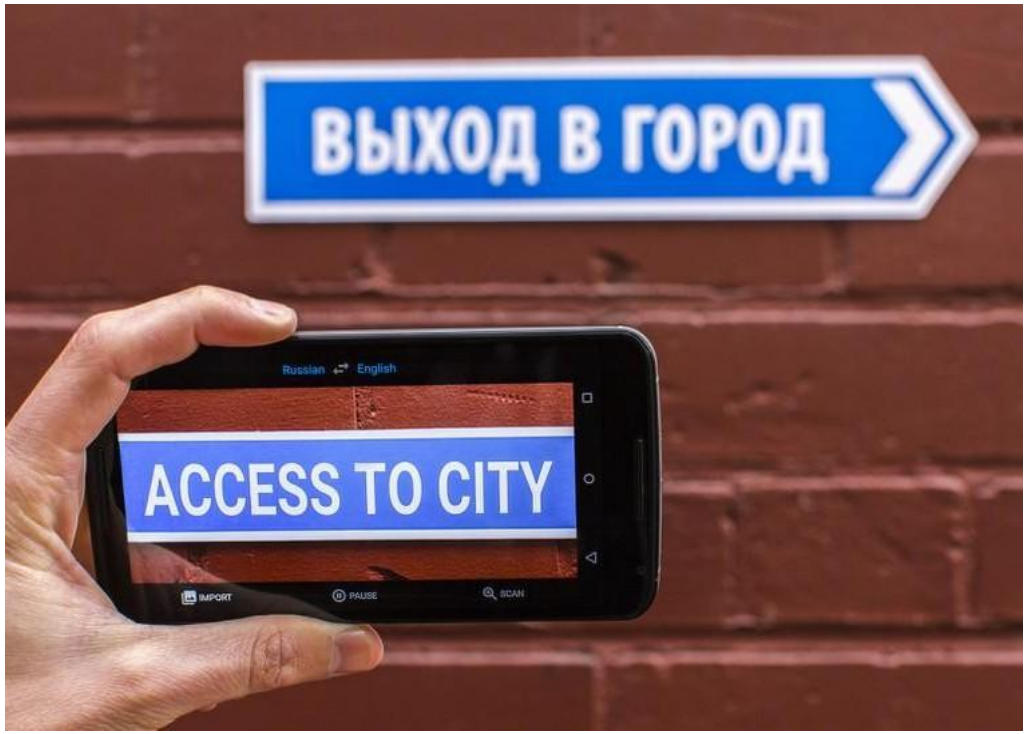




AI assistants, such as **Google Assistant** and **Amazon Alexa**, leverage deep learning to provide a wide range of functionalities, including voice recognition, natural language understanding, and response generation.

Applications of Deep Learning: Tourism





Deep learning technologies are transforming the tourism industry by enabling real-time language translation, personalized travel recommendations, and enhanced customer experiences.

Applications of Deep Learning: Tourism

Real-World Examples

•**Google Lens and Translate**: These apps use OCR and NMT to translate text in real-time by pointing a smartphone camera at signs, menus, and other documents. This functionality is widely used by travelers worldwide to navigate and understand foreign languages more easily.

•**TripAdvisor**: Uses deep learning to personalize content for users, providing tailored recommendations and reviews based on user preferences and past behavior.

•**Kayak and Skyscanner**: These platforms utilize AI to provide dynamic pricing information, personalized travel deals, and optimal booking times, helping users find the best deals for flights, hotels, and car rentals.


E-Commerce


BANKING &
FINANCE

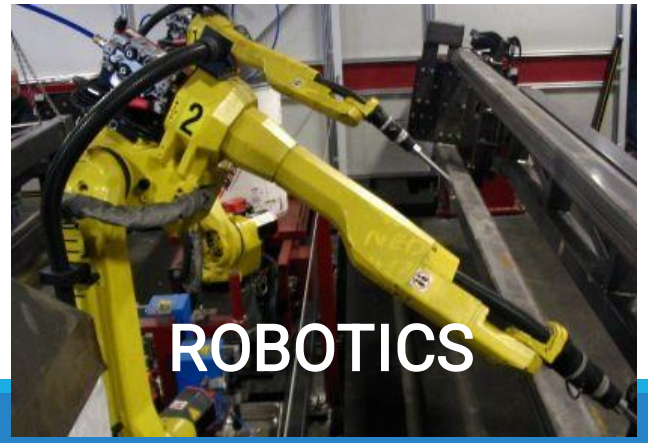

EDUCATION

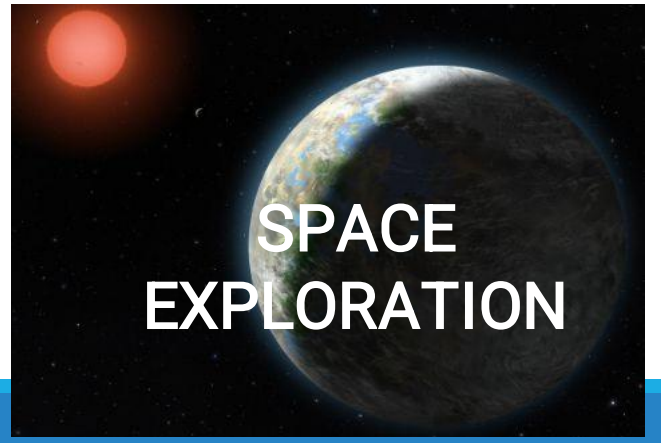

CYBER SECURITY


SPORTS

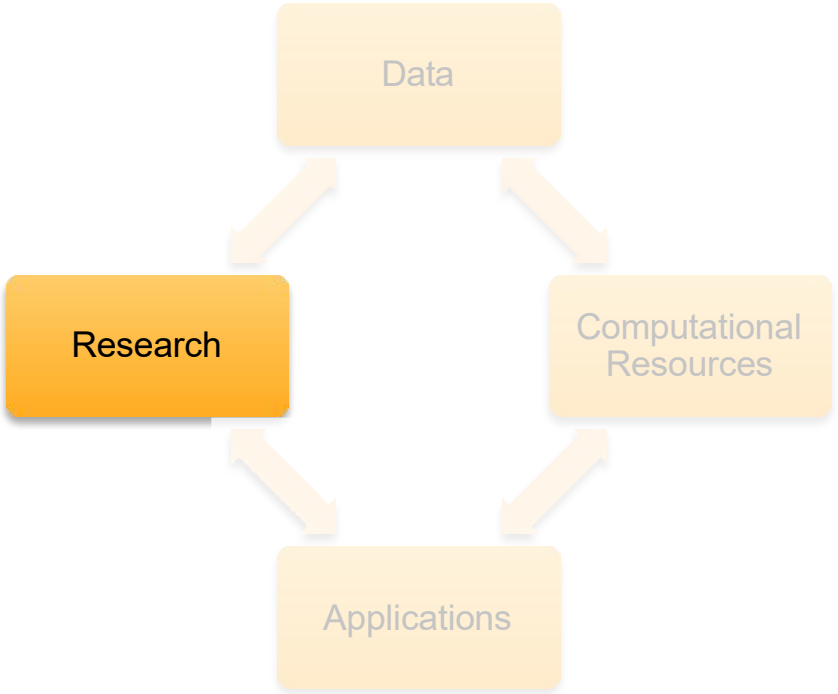

DEFENCE

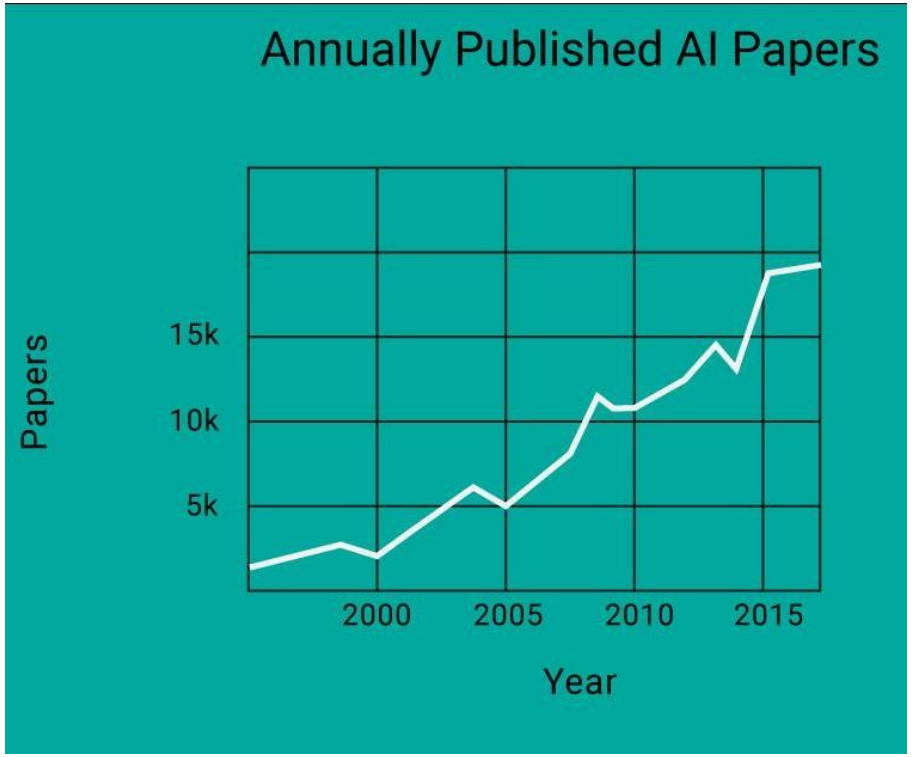

SOCIAL MEDIA


ROBOTICS


SPACE
EXPLORATION

Availability of Data and Computational Resources leads to more Applications and Research





https://ourworldindata.org/grapher/annual-scholarly-publications-on-artificial-intelligence?time=2013..latest&region=Asia&country=USA~GBR~DEU~CHN~OWID_WRL~IND~AUS#explore-the-data

https://hai.stanford.edu/news/ai-index-state-ai-13-charts